

Peiyu (Georgia) Li

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Education

- **University of Notre Dame**, Notre Dame, USA Aug. 2024 - Present
MSc Computer Science and Engineering GPA: 3.93/4.0
Advisor: [Prof. Nitesh V. Chawla](#) (ACM, IEEE, AAAI, AAAS Fellow)
- **Sun Yat-sen University**, Guangzhou, China Aug. 2019 - Jun. 2023
BSc Philosophy, Major in Logic GPA: 3.9/4.0 (92/100) Ranking: Top 4

Working Experience

- **Research Assistant** | [DIAL Lab](#) | University of Notre Dame Oct. 2023 – Present
 - Conducted research in data science, machine learning, and network science for social good.
- **Teaching Assistant: Python in Data Science** | University of Notre Dame Aug. 2024 – Present
 - Supported instruction in Python libraries (Pandas, NumPy, Seaborn) for data analysis and visualization.

Skills

Programming Languages: Proficient in C++, Python; Familiar with Java, SQL, JavaScript

Tools & Platforms: PyTorch, LangChain, NextJS, React, REST, GCP/AWS, Git, Linux, LaTeX, HTML, CSS

Machine Learning: Agent, RAG, Multimodal AI, LLM Alignment Research

Languages: Bilingual in Chinese and English (IELTS: 7.5)

Publications

“**ChefFusion: Multimodal Foundation Model Integrating Recipe and Food Image Generation**”, *CIKM* 2024. DOI: [10.1145/3627673.3679885](https://doi.org/10.1145/3627673.3679885) (First Author)

Research Experience

Code Generation Optimization for Functional Correctness Mar. 2025 – Ongoing

Role: Graduate Research Assistant, Supervisor: Dr. Ting Hua

- Developed a novel optimization method to enhance functional correctness in code generation tasks, targeting improvements in problem-solving capabilities for programming models.
- Introduced a dynamic mechanism to refine model predictions through **iterative preference learning**, focusing on execution success and problem-solving efficiency.
- Testing the approach on various programming benchmarks, including **HumanEval**, **MBPP**, and **APPS**, to assess its effectiveness in solving code generation tasks accurately.

AI-Powered Crochet Pattern Reasoning and Generation

Jan. 2025 – Ongoing

Role: Graduate Research Assistant, Supervisor: Prof. Nitesh V. Chawla

- Developed a **multimodal inference agent** based on **LangChain** and implemented a Retrieval-Augmented Generation (**RAG**) system with optimized database indexing, enhancing the accuracy and stability of complex pattern inference.
- Designed **novel evaluation metrics** and introduced a **new problem formulation** for LLMs in crochet pattern reasoning.
- Curated a specialized dataset to benchmark and advance this research direction.

Role: Graduate Research Assistant, Supervisor: Prof. Nitesh V. Chawla

- Processed the **Recipe1M dataset** (300K+ samples, 3TB) to address **long-context challenges** in multimodal food AI.
- Developed a **multimodal** AI framework using **PyTorch**, integrating text-to-text, text-to-image, and image-to-text tasks, enabling real-time **interactive multimodal dialogue** via a Gradio demo.
- Leveraged **OPT-6.7B** for recipe generation, **Stable Diffusion v1.5** for image synthesis, and **PEFT-based adapters** for **LLM alignment** and efficient cross-modal fusion.

Projects

Embedded Development: AR Smart Glasses with On-Device Personalized LLM (Ongoing)

- Developed a **distributed AI processing pipeline**, integrating **Bluetooth and Wi-Fi communication** to efficiently transfer data between audio capture devices, Raspberry Pi, and AR smart glasses for real-time processing.
- Integrating **Whisper speech recognition** and **AR waveguide display** for real-time AI-driven voice and text interactions.
- Optimizing edge computing performance on **Raspberry Pi** by applying **4-bit/8-bit model quantization**, reducing latency and power consumption.

Scalable Travel Booking Application

- Built a **full-stack travel booking system** with secure authentication, payment processing, and reservation management.
- Developed a robust backend using **Node.js, Express, and MongoDB**, implementing a **RESTful API** for seamless data operations.

User Preference Inference

- Designed a model to infer implicit user needs from LLM dialogues, achieving **first place** in the IEEE North America Student Challenge (Kaggle).
- Leveraged LoRA for lightweight fine-tuning, boosting accuracy from 76.718% to 96.333%.

Modular Tangram Game (C++ Game Development)

- Designed and developed a modular Tangram puzzle game using C++, incorporating advanced features such as **metaprogramming, smart pointers, Curiously Recurring Template Pattern (CRTP), polymorphism, and parallel processing**.
- Integrated **real-time audio feedback and event management** to ensure optimal performance and scalability.

Recipe Search & Management Application

- Created a **dynamic web application** that enables users to search and manage recipes, featuring real-time data retrieval, adjustable serving sizes, and persistent session management.
- Designed a responsive and user-friendly UI using **HTML, CSS, and JavaScript**, ensuring cross-device compatibility and enhanced user experience.
- Implemented **API-based search functionality**, optimized performance for faster loading speeds, and added features like bookmarks and private recipe management to improve user retention.